Bluestock Mutual Fund Analytics

Capstone Project I — Final Report

Performance, Risk, Investor Behaviour and SIP Analytics

27 July 2026 – 11 August 2026

This report covers the data preparation, analysis, risk metrics, investor analysis, recommendation model and Power BI dashboard developed during the project.

1. Executive Summary

The aim of this project was to analyse mutual fund data from different sources and turn it into useful information about fund performance, risk and investor behaviour. The work started with raw CSV files and continued through data cleaning, exploratory analysis, performance calculations and advanced analytics. The final results were then used in a Power BI dashboard.

The project covers CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta and Maximum Drawdown. It also includes Historical VaR and CVaR, rolling 90-day Sharpe analysis, investor cohort analysis, SIP continuity analysis, a simple fund recommender and sector concentration using HHI.

The investor analysis produced 32,499 transaction observations for 2024 and 279 for 2025. There were 19,716 SIP transactions. After selecting investors with at least six SIP transactions, 1,362 investors remained. Using the 35-day average-gap rule, 1,332 were marked at-risk and 30 were not.

The recommendation model was also tested. For a High risk appetite, it returned Mirae Asset Tax Saver Fund, ICICI Pru Midcap Fund and DSP Midcap Fund as the top three based on Sharpe Ratio.

2. Project Objectives

The project was designed to answer both analytical and business questions. Instead of looking only at fund returns, the analysis combines performance, risk, investor behaviour and portfolio structure.

Main objectives

- Clean and organise the available mutual fund datasets.
- Explore industry, fund and investor trends.
- Compare funds using return and risk-adjusted measures.
- Measure downside risk using VaR and CVaR.
- Study how rolling Sharpe changes over time.
- Group investors by their first transaction year.
- Identify possible SIP continuity problems.
- Build a simple risk-based fund recommendation model.
- Measure sector concentration in equity funds.
- Present the results through Power BI.

3. Data Sources

The project uses several datasets covering different parts of the mutual fund ecosystem. The raw files were kept separately from the processed data so that the original inputs were not changed.

Dataset	Use in the project
01_fund_master.csv	Fund details, category, plan, benchmark, expense ratio and risk category.
02_nav_history.csv	Historical NAV values and return calculations.
03_aum_by_fund_house.csv	AUM and fund-house level analysis.
04_monthly_sip_inflows.csv	Monthly SIP inflow trends.
05_category_inflows.csv	Category-wise inflow analysis.
06_industry_folio_count.csv	Industry folio counts.
07_scheme_performance.csv	Scheme performance information.
08_investor_transactions.csv	Investor transactions and SIP activity.
09_portfolio_holdings.csv	Sector weights and holdings for HHI.
10_benchmark_indices.csv	Benchmark data for comparison.

The processed folder contains cleaned NAV history, scheme performance and investor transaction data. The project also contains a data dictionary and SQL assets for the structured part of the workflow.

4. Data Preparation and ETL

The data flow was kept fairly simple. The raw files are the starting point, Python scripts are used for ingestion and cleaning, and the cleaned data is then used by the notebooks. The notebooks produce the performance and advanced analytics outputs, which are saved as CSV files and charts.

Overall flow

Raw data → cleaning and validation → processed data → EDA → performance analysis → advanced analytics → outputs → Power BI

Project folders

• `data/raw` contains the original datasets.

• `data/processed` contains the cleaned datasets used later in the project.

• `notebooks` contains EDA, performance and advanced analytics work.

• `outputs` contains the calculated metrics.

• `reports/charts` contains charts produced during the analysis.

• `reports/dashboard` contains the Power BI dashboard and screenshots.

• `scripts` contains ingestion, cleaning, validation and recommendation code.

• `sql` contains the database/query part of the project.

5. Exploratory Data Analysis

The EDA stage was used to get an overall understanding of the data before moving into the more detailed calculations. The project includes charts for AUM, NAV, folios, categories, geography, investor activity, SIP activity and sector allocation.

Areas covered

• AUM and folio growth were used to look at industry-level changes.

• NAV trends provided the time-series basis for return calculations.

• Category and sector charts helped compare different parts of the market.

• State and city-tier charts were used to understand investor distribution.

• SIP charts showed recurring-investment activity.

• Correlation analysis was used to understand relationships between selected performance variables.

The EDA stage was also useful for checking which datasets were suitable for later analysis. For example, transaction dates were needed for SIP continuity and portfolio sector weights were needed for HHI.

6. Fund Performance Analysis

Several performance measures were calculated instead of relying on one metric. CAGR gives a view of compounded growth, while Sharpe and Sortino give more context about the risk taken to achieve returns. Alpha and Beta were used for benchmark-relative behaviour and Maximum Drawdown was used to understand downside.

Metrics used

• CAGR — compounded annual growth.

• Sharpe Ratio — return compared with overall volatility.

• Sortino Ratio — return compared with downside volatility.

• Alpha — benchmark-relative excess performance.

• Beta — sensitivity to benchmark movement.

• Maximum Drawdown — largest observed fall from a peak.

• Fund Scorecard — combines several ranked measures for comparison.

The outputs were saved separately in the outputs folder, including cagr_report.csv, sharpe_values.csv, sortino_values.csv, alpha_beta.csv, max_drawdown.csv and fund_scorecard.csv.

7. Alpha and Beta Results

The supplied Alpha/Beta output contains 40 funds. Alpha values range from 0.028969 to 0.303370. The highest alpha in the supplied file is SBI Small Cap Fund - Regular Plan - Growth at 0.303370. The lowest is UTI Mid Cap Fund - Regular - Growth at 0.028969.

Highest alpha funds in the supplied output

Fund	Alpha	Beta
SBI Small Cap Fund - Regular Plan - Growth	0.303370	-0.023196
DSP Small Cap Fund - Regular - Growth	0.300579	0.011455
ICICI Pru Midcap Fund - Regular - Growth	0.292636	0.000549
Mirae Asset Tax Saver Fund - Regular - Growth	0.282704	0.018134
Kotak Flexicap Fund - Regular - Growth	0.273305	-0.022830
HDFC Mid-Cap Opportunities Fund - Regular - Growth	0.271954	0.005104
Mirae Asset Large Cap Fund - Regular - Growth	0.269838	0.023684
DSP Midcap Fund - Regular - Growth	0.265986	-0.002523

Alpha should be considered together with Beta and the other risk measures. A higher alpha by itself does not tell us whether a fund is suitable for a particular investor.

8. VaR, CVaR and Rolling Sharpe

Historical VaR was calculated at the 95% level using the 5th percentile of daily returns. CVaR was calculated as the average return below the VaR threshold. These are historical measures, so they describe the observed return distribution rather than guaranteeing a future loss limit.

A rolling 90-day Sharpe Ratio was calculated for five key funds. The calculation uses the rolling mean return divided by rolling standard deviation and annualises the result using √252. This gives a time-based view of whether risk-adjusted performance is improving or weakening.

Main files produced

• `outputs/var_cvar_report.csv`

• `reports/charts/rolling_sharpe_chart.png`

• `outputs/sharpe_values.csv`

9. Investor Cohort Analysis

Investors were grouped according to the year of their first transaction. The analysis produced 32,499 transaction observations for 2024 and 279 for 2025.

Cohort	Transactions	Average amount	Total invested
2024	32,499	107,422.54	3,491,125,187
2025	279	109,158.58	30,455,243

The 2024 cohort is much larger in the available data and also has a much higher total invested amount. The cohort output also records fund preference for each year, which can be used to compare the schemes most frequently chosen by the two groups.

10. SIP Continuity Analysis

The SIP analysis started with 19,716 SIP transactions. Investors with at least six SIP transactions were then selected, giving 1,362 eligible investors. For each investor, the dates were ordered and the average gap between consecutive SIP transactions was calculated.

An investor was marked **at-risk** when the average gap was greater than 35 days. The final result was 1,332 at-risk investors and 30 investors who were not flagged.

Interpretation

This should be treated as a monitoring signal rather than a prediction. A long gap can happen for many reasons, and the analysis does not establish why an investor missed a SIP.

11. Fund Recommendation Model

The recommender is deliberately simple. The user enters Low, Moderate or High risk appetite. The program matches that input to the fund's risk category and then sorts the matching funds by Sharpe Ratio.

High-risk test

Rank	Fund	Sharpe
1	Mirae Asset Tax Saver Fund - Regular - Growth	1.234930
2	ICICI Pru Midcap Fund - Regular - Growth	1.180101
3	DSP Midcap Fund - Regular - Growth	1.132122

The model was implemented in `scripts/recommender.py` and tested from the terminal. It is a screening tool rather than financial advice and would need more investor information for a production version.

12. Sector Concentration — HHI

Portfolio holdings were used to calculate the Herfindahl-Hirschman Index. The calculation uses the sector weight for each fund and sums the squared decimal weights. A higher HHI means that more of the portfolio is concentrated in fewer sectors.

Sample completed results

AMFI code	HHI
100016	0.139534
100033	0.147592
101206	0.129332
101207	0.200700
102885	0.174709

The HHI result is useful when a fund looks strong on returns but has exposure concentrated in a small number of sectors. The completed output is stored as `outputs/sector_hhi.csv`, with the chart saved under `reports/charts/`.

13. Power BI Dashboard

The Power BI dashboard is the final visual layer of the project. It brings the results from the analytical files into a format that can be explored using charts, filters and slicers.

Dashboard pages

• Industry Overview

• Fund Performance

• Investor Analytics

• SIP & Market Trends

The project folder also contains the PBIX file and four dashboard screenshots. These should be included in the final repository because they provide visual evidence of the completed dashboard.

14. Key Findings and Recommendations

Key findings

• The 2024 cohort is much larger than the 2025 cohort in the available transaction data.

• The 2024 cohort also accounts for the larger total invested amount.

• 19,716 SIP transactions were analysed.

• 1,362 investors met the 6+ SIP condition.

• 1,332 of those investors were flagged at-risk using the 35-day average-gap rule.

• Alpha/Beta analysis was completed for 40 funds.

• The High-risk recommender returned three funds with Sharpe Ratios above 1.13.

• HHI adds a useful diversification measure alongside the return and risk metrics.

Recommendations

• Compare funds using both return and risk-adjusted measures instead of CAGR alone.

• Use SIP gap flags to identify investors who may need follow-up.

• Look at sector concentration when comparing otherwise similar equity funds.

• Keep the recommender transparent and add suitability information before using it in a real investor-facing system.

• Automate data refresh and dashboard updates if the project is extended.

15. Limitations and Conclusion

Limitations

• The analysis uses the supplied historical datasets and does not predict future performance.

• VaR and CVaR depend on the historical return distribution and may not represent future extreme events.

• The 35-day SIP threshold is a project-defined rule and should be validated before operational use.

• The recommender only uses risk category and Sharpe Ratio and is not a personalised investment recommendation.

• Cohort results describe the available transaction data and should not automatically be treated as the complete market population.

Conclusion

The project brings together fund performance, risk, investor behaviour and portfolio concentration in one workflow. Looking at these areas together gives a better picture than looking at returns alone. The Power BI dashboard then makes the results easier to compare and explore.

Overall, the completed work provides a practical base for further development. The next step would be to automate the full pipeline, improve the recommendation logic and connect the dashboard to a regularly refreshed data source.

16. Project Outputs and Submission Checklist

• Advanced_Analytics.ipynb — advanced risk, cohort, SIP and HHI analysis.

• var_cvar_report.csv — Historical VaR and CVaR results.

• rolling_sharpe_chart.png — rolling 90-day Sharpe visual.

• cohort_analysis.csv — investor cohort results.

• sip_continuity.csv — SIP gap and at-risk classification.

• recommender.py — risk-based fund recommendation model.

• sector_hhi.csv — fund-level sector concentration.

• sector_hhi_chart.png — HHI visual.

• Power BI PBIX and dashboard screenshots.

• Final report and presentation.

The project can now be packaged for the final GitHub release after a quick check of the report, presentation and repository files.

Appendix — Supplied Alpha/Beta Output

The complete 40-fund Alpha/Beta CSV used for the report is provided separately as `alpha_beta_final.csv`.

End of report